

Course Handout



Sem-I M 2026: CS1001-Introduction to Computing (L-T-P-C: 3-0-2-4)

Course Objectives: This course is designed to develop a strong foundation in computers and programming through fundamental problem-solving techniques. It aims to nurture logical thinking in students using a learning-by-doing approach with the C programming language. The course includes a hands-on lab component to provide practical experience in implementing core programming concepts and solving real-world problems using C.

Course Outcomes:

After completion of the course, students will be able to

CO1: Describe and explain foundation concepts of Computing, Computer science and identify situations where computational methods are useful

CO2: Outline and describe the various programming fundamentals and its constructs

CO3: Analyze, describe and apply a programming language syntax by example of C language

CO4: Learn, code, and develop computer programs using the programming language

CO5: Edit, compile, debug, correct, recompile, and run a program

CO6: Recognize possible errors during program execution

Syllabus:

M1 [6 Hours]: Introduction to Computers and Programming (CO1)

- Introduction to C, Structure of a C program, Low-level vs High-level languages.
- **Number System:** Binary arithmetic, Conversion from one base to another, Representation of negative and floating-point numbers, IEEE standards, and history.

M2 [6 Hours]: C Fundamentals & Basic Syntax (CO2)

- Overview of C, Variables, Data Types, and Constants.
- Operators & Expressions (Arithmetic, Relational, Logical, Bitwise).
- Basic I/O (other than file I/O).

M3 [8 Hours]: Control Flow & Arrays (CO3)

- **Decision making and Branching:** Conditions-if-else constructs, switch case, Ternary conditional statements.
- **Decision making and looping:** for loop, while loop, do-while, nested loops, break, continue.
- **Arrays:** 1D (one dimension) arrays, Multidimensional arrays, Character arrays, and Strings.

M4 [8 Hours]: User-Defined Functions (CO4, CO5, CO6)

- **Functions:** Definition of function in C, call by value, functions with arrays.
- **Recursive functions:** Types of recursions - direct recursion, indirect recursion, Tail recursion.

M5 [8 Hours]: Pointers and Composite Data Types (CO4, CO5, CO6)

- **Pointers:** Basics, pointer arithmetic, pointers vs. arrays.
- **Dynamic Memory:** malloc, realloc, calloc, free.
- **Structures & Unions:** Definition, nesting, pointers to structs, array of structures.

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- **Passing Variables:** Call by value, call by reference, passing arrays and pointers.

M6 [6 Hours]: File Handling & Advanced Topics (CO4, CO5, CO6)

- **File Handling:** File Opening Modes, fopen, fread, fwrite, fclose.
- **Advanced Topics:** Command line arguments, debugging (gdb, assert()).

Textbook:

T1: BW Kernighan, DM Ritchie, *The C programming language*, 2nd Ed, India: Prentice Hall, 1988.

T2: Stephen Prata, *C primer plus*, 6th Ed, USA: Addison-Wesley Professional, 2014.

Reference Books:

R1: RG Dromey, *How to Solve it by Computer*, 1st Ed, Pearson, 2006.

R2: Yaswanth Kanethkar, *Let us C*, BPB Publications, 16th Ed, 2017.

R3: E Balaguruswamy, *Programming in ANSI C*, McGraw-Hill, 8th Ed, 2019.

R4: MD Harvey and PJ Deitel, *C: how to program*, Prentice-Hall, Inc., 1992.

LAB ASSIGNMENTS:

- All Lab assignments will be done in a Unix-like shell environment.
- Each lab participation is mandatory for evaluation and award marks.

Evaluations:

Component	Weightage (%)	Date & Time	Nature of Component
Minor-I	15%	Academic Calendar	Closed Book
Minor-II	15%	Academic Calendar	Closed Book
End Term	40%	Academic Calendar	Closed Book
Lab Participation (10) (Continuous Evolution)	10%	Each Lab Hrs	Open Book
Lab Exams (2)	20%	A week before will be announced in the class	Closed book

Make-up Policy: It is as per the university norms.

Course Notices will be communicated either by email or in class.

Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all students throughout the semester, and any type of academic dishonesty is not acceptable.